This assignment will not be graded and is only for practice.

Level 1

1) Consider two vectors $a = (1, -2, 2), b = (4, 0, -3)$. Using the standard dot product as the inner product, the projection of a in the direction of b is given by ***1 point***

- ( ) $(\frac{-8}{25}, 0, \frac{6}{25})$
- ( ) $(\frac{-8}{25}, \frac{6}{25})$
- ( ) $(\frac{8}{25}, 0, \frac{-6}{25})$
- ( ) $(\frac{8}{25}, \frac{-6}{25})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(\frac{-8}{25}, 0, \frac{6}{25})$

2) Consider an orthonormal basis $\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\}$ of $\mathbb{R}^2$. If $(3, 5) = c_1v_1 + c_2v_2$ where $c_1, c_2 \in \mathbb{R}$ and $v_1 = \frac{1}{\sqrt{5}}(1, 2), v_2 = \frac{1}{\sqrt{5}}(-2, 1)$, then choose the correct option. ***1 point***

- ( ) $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$
- ( ) $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$
- ( ) $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$
- ( ) $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$

3) Consider an orthonormal basis $\{(1, 0, 0), (0, 1, 0)\}$ for a subspace W in $\mathbb{R}^3$. If $x = (1, 2, 3)$ is a vector in $\mathbb{R}^3$, then which of the following represents a vector in W whose distance from x is the least? Consider dot product as the standard inner product. ***1 point***

- $(2, 4, 0)$
- $(3, 4, 0)$
- $(4, 5, 0)$
- $(1, 2, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2, 0)$

Level 2

Consider the inner product $\langle a, b\rangle = a_1b_1 - a_1b_2 - a_2b_1 + 4a_2b_2$ where $a = (a_1, a_2), b = (b_1, b_2)$ are vectors in $\mathbb{R}^2$. Use this information to answer questions 4 and 5.

4) Let $x = (1, 2)$. Find the projection of x in the direction of $(3, 4)$ using the inner product defined above.

1 point

- $\frac{25}{49}(3, 4)$
- $\frac{11}{25}(3, 4)$
- $\frac{25}{49}(1, 2)$
- $\frac{11}{25}(1, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{25}{49}(3, 4)$

5) Let $x = (1, 2)$, find the projection of x in a direction perpendicular to $(3, 4)$.

1 point

- $\frac{25}{49}(-4, 3)$
- $\frac{11}{25}(4, -3)$
- $(-\frac{26}{49}, -\frac{2}{49})$
- $(\frac{26}{49}, -\frac{2}{49})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-\frac{26}{49}, -\frac{2}{49})$

6) Consider an orthogonal basis $\{(1,2,1),(-2,0,2)\}$ of a subspace W, of the inner product space $\mathbb{R}^3$ with respect to the dot product. If $y = (1,2,3) \in \mathbb{R}^3$, then find $Proj_W(y)$. ***1 point***

- $(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$
- $(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$
- $(\frac{1}{3}, -\frac{8}{3}, \frac{7}{3})$
- $(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$

Your score is: 0/6